Thread Report

1. Thread scope

- Configure and extend a Notion database ("Daily Health Metrics") used in an Apple Health → Notion automation pipeline, specifically:
  - Add several formula-based flag and category properties via the Notion API.
  - Normalize select options (with emoji) for recovery/booster decisions.
  - Generate ready-to-run curl commands for schema updates and a test row to validate formulas.
  - Clarify how to represent SpO_2 values (format and thresholds).

2. Confirmed facts

- The Notion database "Daily Health Metrics" exists with ID `339d7cd8-531f-819f-85b2-c769696ea27c` and is used for daily health metrics.
- The database already included these working formula properties (verified via UI):
  - `day_of_week`
  - `sleep_efficiency_pct`
  - `hr_dip_pct`
  - `hr_dip_category`
  - `flag_deep_sleep_low`
- The database schema contains many numeric and select fields corresponding to sleep, HR, HRV, SpO_2, RHR, workouts, and context (e.g., `sleep_deep_min`, `hrv_sdnn_ms`, `rhr_bpm`, `spo2_min_pct`, `resp_rate_brpm`, `sleep_awakenings_count`, `sleep_longest_wake_min`, `sleep_waketime_num`).
- `hr_dip_category` is a formula property, not a select.
- `booster_decision` is a select property with three options initially named without emoji ("Green light", "Borderline - first only", "Recovery day - skip").
- `spo2_min_pct` and `spo2_avg_pct` are configured as Notion number fields with `percent` format.
- A dedicated test page titled "⊠ Formula Test Row – DELETE ME" was created in the database to exercise the formulas.
- On the test row, `hr_dip_pct` computed as 19.4 and `hr_dip_category` showed "Normal".
- On the test row, `sleep_efficiency_pct` computed as 81.

- On the test row, `spo2_min_pct` displayed as `8800%` when the value `88` was entered, due to percent formatting.

3. Attempts made

- Designed and provided curl `PATCH /v1/databases/{id}` commands to add the following formula properties:
  - `flag_deep_gate_50`
  - `flag_hrv_very_low`
  - `flag_rhr_elevated`
  - `flag_spo2_low`
  - `flag_resp_rate_high`
  - `flag_sleep_fragmented`
  - `flag_early_wake`
  - `flag_hr_dip_non_dipping`
  - `flag_recovery_red_gate`
  - `recovery_category`
- Provided a curl `PATCH` command to update `booster_decision` select options to include emoji:
  - `☒ Green light`
  - `⚠ Borderline - first only`
  - `☒ Recovery day - skip`
- Provided a curl `POST /v1/pages` command to create a test row with concrete numeric values for sleep, HR, HRV, RHR, SpO_2, and respiratory rate, intended to trigger most flags.
- Proposed a formula for `flag_recovery_red_gate` based on an incomplete specification from the user, filling in the missing logic.
- Proposed a formula for `recovery_category` that derives an overall color-coded category (GREEN/YELLOW/RED) from the boolean flags.
- After observing `8800%` for SpO_2, suggested changing the field format of `spo2_min_pct` (and by extension `spo2_avg_pct`) from `percent` to plain `number` to avoid the scaling issue.
- Discussed the alternative of keeping `percent` format and instead using decimal values and thresholds (e.g., `0.90` vs `90`), but framed this as less convenient.

4. What failed or proved fragile

- Using `percent` format for `spo2_min_pct` while entering whole-number percentages (e.g., `88` for 88%) proved fragile:
  - Notion multiplied the stored value by 100 for display, showing `8800%` instead of `88%`.
  - This mismatch created confusion and required either a schema change or a different data-entry convention.

- The original user specification for `flag_recovery_red_gate` was truncated ("or"), leaving the exact intended logic ambiguous; any filled-in logic is therefore speculative rather than fully confirmed.
- There is an unresolved subtlety around SpO_2 thresholds and whether formulas should compare against `< 90` (whole-number representation) or `< 0.90` (decimal representation).

5. What appeared to work

- All newly added flag formulas appeared to compute correctly on the test row:
  - `flag_deep_sleep_low` = true (28 < 35).
  - `flag_deep_gate_50` = true (28 < 50).
  - `flag_hrv_very_low` = true (32 < 40).
  - `flag_rhr_elevated` = true (72 > 68).
  - `flag_spo2_low` = true, given the `< 90` logic and entered value of 88 (despite the percent-display issue).
  - `flag_resp_rate_high` = true (19 > 18).
  - `flag_sleep_fragmented` = true (6 awakenings and longest wake 20 > 15).
  - `flag_early_wake` = true (sleep_waketime_num 360 within 330–445).
  - `flag_hr_dip_non_dipping` did **not** fire, consistent with `hr_dip_pct` = 19.4 ≥ 10, so the logic worked as intended.
  - `flag_recovery_red_gate` = true on that row, consistent with low HRV and low deep sleep and/or elevated RHR / low SpO_2.
- `hr_dip_pct` and `hr_dip_category` formulas continued to behave as expected after adding dependent flags.
- The test row pattern (single synthetic row with extreme values) proved useful for validating multiple formulas in one place.

6. Recommendations made in this thread

1. **Represent SpO_2 as a plain number (not percent format) and compare against `< 90` for a 90% threshold.**
   - Rationale: avoids the 8800% display issue and matches how devices present SpO_2 (whole percent values).
   - Status: **Still plausible.** This is consistent with the observed behavior and with typical health-tracking UX; no contradiction appears in the thread.
2. **Alternatively, keep SpO_2 as `percent` format but use decimal values (0.88) and thresholds `< 0.90`.**
   - Rationale: aligns mathematically with a percent field, but awkward for data entry.
   - Status: **Still plausible but discouraged.** The thread itself frames this as less natural, not as an error.
3. **Use the following formulas for flags (directly from the thread):**

- `flag_deep_gate_50:prop("sleep_deep_min") < 50`
- `flag_hrv_very_low:prop("hrv_sdnn_ms") < 40`
- `flag_rhr_elevated:prop("rhr_bpm") > 68`
- `flag_spo2_low:prop("spo2_min_pct") < 90` (assuming plain number representation)
- `flag_resp_rate_high:prop("resp_rate_brpm") > 18`
- `flag_sleep_fragmented:prop("sleep_awakenings_count") >= 5 or prop("sleep_longest_wake_min") > 15`
- `flag_early_wake:prop("sleep_waketime_num") >= 330 and prop("sleep_waketime_num") <= 445`
- `flag_hr_dip_non_dipping:prop("hr_dip_pct") < 10`
- Status: **Still plausible.** These formulas behaved as expected in the test row, and no contradictions were raised.

4. **Use a `flag_recovery_red_gate` formula combining HRV, deep sleep, RHR, and SpO_2 red flags:**
   - Proposed expression: `(prop("hrv_sdnn_ms") < 40 and prop("sleep_deep_min") < 35) or prop("rhr_bpm") > 68 or prop("spo2_min_pct") < 90`
   - Status: **Unclear / partially speculative.**
     - The user's original description for this gate was truncated, so the exact intended logic is not fully documented.
     - The formula was inferred to match "red gate" semantics but cannot be confirmed as the user's original intent.
5. **Derive `recovery_category` as a text formula from flags:**
   - Proposed structure:
     - If `flag_recovery_red_gate` → `"⛔ RED"`
     - Else if any of `flag_deep_gate_50`, `flag_sleep_fragmented`, `flag_early_wake` → `"⚠ YELLOW"`
     - Else → `"✅ GREEN"`
   - Status: **Unclear / speculative.**
     - The exact weighting of flags for YELLOW vs RED is a design decision; the thread documents one version but not as an explicitly finalized clinical rule.
6. **Rename `booster_decision` options to emoji-prefixed labels:**
   - `✅ Green light` / `⚠ Borderline - first only` / `⛔ Recovery day - skip`.
   - Status: **Still plausible.**
     - This is a UX/labeling decision, and no conflicts are documented.
     - The thread explicitly warns that renaming select options can orphan old values, which remains a valid caution.

7. Artifacts worth preserving

- **Property names and intended formulas:**
  - Flag fields:
    - `flag_deep_sleep_low` (existing): `prop("sleep_deep_min") < 35`.
    - `flag_deep_gate_50`: `prop("sleep_deep_min") < 50`.
    - `flag_hrv_very_low`: `prop("hrv_sdnn_ms") < 40`.
    - `flag_rhr_elevated`: `prop("rhr_bpm") > 68`.
    - `flag_spo2_low`: `prop("spo2_min_pct") < 90` (assuming plain number).
    - `flag_resp_rate_high`: `prop("resp_rate_brpm") > 18`.
    - `flag_sleep_fragmented`: `prop("sleep_awakenings_count") >= 5 or prop("sleep_longest_wake_min") > 15`.
    - `flag_early_wake`: `prop("sleep_waketime_num") >= 330 and prop("sleep_waketime_num") <= 445`.
    - `flag_hr_dip_non_dipping`: `prop("hr_dip_pct") < 10`.
    - `flag_recovery_red_gate`: speculative but documented composite gate combining HRV, deep sleep, RHR, and SpO_2.
  - Category fields:
    - `recovery_category`: formula mapping flags to `"⊠ GREEN"`, `"⊠ YELLOW"`, `"⊠ RED"`.
- **Select labels:**
  - `booster_decision` recommended options with emoji.
- **Test-row pattern:**
  - Use a dedicated "Formula Test Row – DELETE ME" with intentionally "bad" values to validate formula logic.
- **Schema pattern:**
  - Use raw numeric fields for physiological inputs and layered formulas (percentages, flags, categories) on top, referencing other formula fields where needed (`hr_dip_pct`, `hr_dip_category`).

8. Open questions left unresolved

- Exact, fully-agreed logic for `flag_recovery_red_gate` (the original specification was truncated, and the final clinical intention is not fully documented).
- Exact, agreed logic for `recovery_category` (which flags should control RED vs YELLOW vs GREEN, and whether additional factors like HR dip should be included).
- Final decision on whether `spo2_min_pct` and `spo2_avg_pct` will be stored as plain numbers or kept as percent-formatted numbers in the long term.
- Whether any other flags or categories should depend on HR baseline or HRV baseline fields (e.g., relative deviations vs absolute thresholds), which is not discussed here.

- How this Notion configuration is integrated with the upstream Apple Health → Notion automation pipeline (the thread assumes it exists but does not document it).

9. Usefulness score

Score: 4 / 5

- Strengths:
  - Provides concrete property names and formula expressions that clearly worked on a real test row.
  - Surfaces an important SpO_2-formatting pitfall and discusses alternatives.
  - Documents a consistent flag-and-category schema pattern that can be reused.
- Limitations:
  - Some key logic (`flag_recovery_red_gate`, `recovery_category`) is necessarily speculative due to the truncated original requirement.
  - Integration details with the broader automation pipeline are not covered.

10. Repo-ready summary

This thread documents a focused configuration pass on the "Daily Health Metrics" Notion database used in the Apple Health → Notion automation project. It establishes multiple boolean flag formulas (deep sleep, HRV, RHR, SpO_2, respiratory rate, sleep fragmentation, early wake, and HR-dipping) and proposes a composite recovery gate and recovery-category logic. A synthetic "Formula Test Row" was used to validate the formulas, and the thread highlights a key schema pitfall: using percent-formatted fields for SpO_2 while entering whole-number percentages causes misleading display values (e.g., 88 → 8800%), leading to the recommendation to store SpO_2 as a plain number and compare against a 90% threshold. Some aspects of the red-gate and recovery-category formulas remain partially speculative because the original specification was truncated, and should be revisited before final clinical interpretation.